



SIMATS

ENGINEERING

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ASSESSMENT TOOL

3

**INTERNET PROGRAMMING
CSA43**

COURSE NAME:

COURSE CODE:

COURSE OUTCOME COVERED

CO2-AT3

CO-2 : Apply CSS

layout and styling techniques to create visually appealing and responsive web interfaces, and use JavaScript to enable interactive client-side behavior. (L3)

Assessment Tool : Short Answer

Test

Weightage: 10%

Total points

0/0



- 1. Question
(5 Marks)

Suppose that a company wants its website to work seamlessly on mobile, tablet, and desktop devices.

- Explain the concept of responsive web design and its importance.
- Describe the key CSS techniques used to achieve responsiveness and explain how CSS Flexbox or Grid can be used to design adaptive layouts.
- Illustrate your answer with a simple example layout and suggest improvements to enhance user experience across different devices.



1) Responsive Web Design (RWD) is a technique that allows a website to automatically adjust its layout according to different screen sizes such as mobile, tablet, and desktop. It improves user experience, accessibility, and SEO while reducing maintenance costs. CSS techniques like media queries, flexible units (%), responsive images, and the viewport meta tag help achieve responsiveness. CSS Flexbox and Grid create adaptive layouts by automatically arranging content into different columns based on screen size. User experience can be improved by optimizing images, using touch-friendly buttons, and maintaining readable font sizes.

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B. Harishwar Reddy (192311338)

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2. Question (5 Marks)

Suppose that a webpage contains a form where user input must be validated before submission. Explain the concept of client-side validation and its purpose.

Describe how JavaScript event handling is used in validation and explain the role of regular expressions in validating inputs such as email or phone numbers. Provide a suitable example and suggest improvements for better usability and security.

2) Client-side validation checks user input in the browser before submitting data to the server, ensuring that only valid information is entered. It improves user experience by providing instant feedback and reducing unnecessary server requests. JavaScript uses events such as onsubmit, oninput, and onblur to perform validation, while Regular Expressions (Regex) verify formats like email addresses and phone numbers. For example, Regex ensures an email follows the correct pattern and a phone number contains exactly 10 digits. Better usability and security can be achieved by displaying clear error messages and combining client-side validation with server-side validation.

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Question 3

(5 Marks)

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Suppose that a webpage layout breaks when the browser window is resized.

Explain the CSS box model and its components, and describe how CSS positioning techniques affect layout structure.

Identify possible causes of layout breaking and explain how media queries can be used to resolve these issues. Suggest best practices to ensure a stable and responsive layout.

3) The CSS Box Model consists of content, padding, border, and margin, which determine the size and spacing of HTML elements. CSS positioning methods such as static, relative, absolute, fixed, and sticky control how elements are placed on a webpage. Layouts may break due to fixed widths, large images, incorrect positioning, or missing responsive styles. Media queries solve these issues by applying different CSS rules for different screen sizes. Using Flexbox/Grid, relative units, optimized images, and responsive testing ensures a stable and responsive webpage.

Question 4

(5 Marks)

Suppose that a website requires dynamic content updates without reloading the entire page.

Explain the concept of the Document Object Model (DOM) and its role in web development.

Describe how JavaScript can be used to manipulate the DOM for updating webpage content dynamically. Provide an example and suggest improvements to enhance performance and user experience.



4) The Document Object Model (DOM) is a tree-like representation of an HTML document that allows JavaScript to access and modify webpage elements dynamically. It enables developers to update text, styles, images, and other content without reloading the page. JavaScript methods such as getElementById(), querySelector(), and innerHTML are commonly used for DOM manipulation. For example, clicking a button can instantly change the text displayed on a webpage. Performance can be improved by minimizing DOM updates, using event delegation, and loading data asynchronously with AJAX or the Fetch API.

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Code of Conduct:

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

B. Harishwar Reddy

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RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	30	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	25	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	20	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	15	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	10	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand

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